

VIGNESH V

Pondicherry | +91 81247 95610 | VigneshV26@outlook.com
github.com/Vignesh-V2000 | linkedin.com/in/vignesh-v26

Professional Summary

MCA graduate with a 6-month internship at NCPOR (Ministry of Earth Sciences, Govt. of India), contributing to AI-powered polar data management systems and neural network-based fog prediction. Skilled in Python, Machine Learning, IoT, and Data Analytics, with a strong interest in AI-driven solutions and software development.

Education

Pondicherry University <i>Master of Computer Applications (MCA)</i>	2024 – 2026
Indira Gandhi College of Arts and Science <i>Bachelor of Computer Applications (BCA)</i>	2021 – 2024

Technical Skills

Languages	Python, Java, C, C++, C#
Web & Frameworks	Django, HTML5, CSS3, JavaScript
AI / ML	Machine Learning, Neural Networks, NLP, scikit-learn
AI Tools	ChatGPT, Gemini, Claude, Perplexity, Hugging Face, Ollama
IoT	ESP32, ThingSpeak, Sensor Integration
Databases	SQLite, PostgreSQL
Tools	Git, GitHub, VS Code, Figma
Design	UI/UX Design

Internship

National Centre for Polar and Ocean Research (NCPOR) <i>Intern – Polar Data Service Section, Ministry of Earth Sciences, Govt. of India</i>	Dec 2025 – May 2026
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- Developed and integrated four AI modules (Intelligent Search, Chatbot, Submission Helper, and Login Messaging) for the National Polar Data Centre (NPDC) portal.
- Implemented fog classification and prediction models using neural networks on polar observation datasets as part of the “AI-Driven Polar Data Management System” project.
- Built a full-stack, Django-based manuscript submission workflow with role-based access control for submitters, reviewers, and data providers.
- Designed and integrated smart caching mechanisms to optimize dataset download performance for the data portal.

Projects

AI-Driven Polar Data Management System (NPDC) <i>Python, Django, Neural Networks, NLP</i>	Dec 2025 – May 2026
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- Engineered an intelligent search module with NLP-based query understanding and relevance ranking for the NPDC data catalogue.
- Built a domain-specific chatbot to assist researchers in navigating polar datasets and submission procedures.
- Developed fog classification models using neural networks trained on Antarctic and Arctic observation data.
- Created automated submission helper tools that validate and pre-process research data uploads.

Journal Manuscript Submission System <i>Python, Django, HTML/CSS, JavaScript, SQLite</i>	Apr 2026
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- Developed a web-based manuscript submission system with a multi-stage review workflow.

- Implemented role-based authentication and dashboards for Submitters, Reviewers, and Data Providers.
- Built responsive user interfaces and administrative tools for reviewer assignment, submission validation, and data management.

Indoor Air Quality Monitor with Predictive Analytics

Jul 2025 – Nov 2025

ESP32, Python, ThingSpeak, scikit-learn

- Built an IoT-based Indoor Air Quality Monitor that tracks PM2.5, temperature, humidity, and CO₂ using an ESP32 microcontroller.
- Integrated real-time data transmission to ThingSpeak for continuous cloud-based monitoring and visualization.
- Applied machine learning (scikit-learn) to predict upcoming air quality trends, enabling proactive environment management.

Trainings & Certifications

- **The Joy of Computing Using Python** – NPTEL (Jan 2025 – Apr 2025)
- **Data Analytics With Python** – NPTEL (Jan 2025 – Apr 2025)

Extra Curricular Activities & Languages

- **Class Representative (MCA):** Served as the primary liaison between students and faculty, coordinating academic activities and organizing department events.
- **Technical Paper Presentation:** Presented a paper on “IoT-based Air Quality Monitoring” at a university-level seminar.
- **Languages:** English (Fluent), Tamil (Native)